MIL-B-11818D
17 December 85
SUPERSEDING
MIL-B-11818C(EL)
10 March 1969

MILITARY SPECIFICATION

BAROMETER, ANEROID ML-102( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of pressure indicating barometer designated as Barometer, Aneroid ML-102( ).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder, Tin Alloy, Tin-Lead Alloy, And Lead Alloy.
MMM-A-181 - Adhesive, Phenol, Resorcinol, Or Melamine Base.

MILITARY

MIL-M-14 - Molding Plastics And Molded Plastic Parts, Thermosetting.
MIL-P-116 - Preservation, Methods Of.
MIL-P-5425 - Plastic, Sheet, Acrylic, Heat Resistant.
MIL-M-13231 - Marking Of Electronic Items.
MIL-F-14072 - Finishes For Ground Electronic Equipment.
MIL-E-55585 - Electronics Equipment And Parts, Packaging Of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, Fort Monmouth, New Jersey 07703-5304, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6660

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards For Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-454 - Standard General Requirements For Electronic Equipment.
MIL-STD-726 - Packaging Requirements Code.
MIL-STD-810 - Environmental Test Methods.
MS 35338 - Washer, Lock-Spring, Helical, Regular (Medium) Series.

2.1.2 Other government documents. The following government drawings form a part of this specification to the extent specified herein:

DRAWINGS

ELECTRONICS COMMAND

SC-A-46420 - Fasteners.
SC-A-47794 - Wood Guide (High Density).
SC-C-20582 - Dial-Barometer ML-102( ).
SC-DL-282023 - Case CY-1059( )/UM.
SM-C-282049 - Barometer, Aneroid ML-102G.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The Barometer, Aneroid ML-102( ) consists of the aneroid mechanism, a mounting frame assembly, an instrument case, and case CY-1059( )/UM. The equipment is designed to measure and indicate the atmospheric pressure in the range from 745 to 1065 millibars(mb).

3.2 First article. When specified in the contract, samples shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. The material shall be as shown on the applicable drawings and as specified herein. However, when a definite material is not specified, a material shall be used which will enable the equipment to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

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3.3.1 Recycled, virgin and reclaimed material. It is encouraged that recycled or reclaimed material, in lieu of virgin material, be used when practical, provided it meets the requirements of this specification.

3.3.2 Adhesives. Wood-to-wood joints shall be made with glue conforming to MMM-A-181, type I or II.

3.3.3 Plastic material. Where not machined, plastic material shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be made smooth or polished.

3.3.4 Wood. Solid wood shall be in accordance with Drawing SC-A-47794.

3.3.5 Cleaning. The equipment shall be cleaned in accordance with requirement 9 of MIL-STD-454.

3.4 Design.

3.4.1 Screws, other fastening devices, and related parts.

3.4.1.1 Form and fit. Screw threads shall be in accordance with FED-STD-H28. Unified thread form and series shall be class 2 fit unless otherwise specified herein.

3.4.1.2 Machine and cap screws, and washers. Machine screws, and cap screws and washers shall be selected from Drawing SC-A-46420 unless otherwise specified herein.

3.4.1.2.1 Machine screws. Machine screws shall be not less than No. 6 nominal size .138 inch (3.51mm), and shall have drilled fillister or pan heads. Flat-head screws shall not be used in sheet or thin material unless the thickness is more than 1.5 times the height of the screw head. The screw head shall be completely seated in the material.

3.4.1.2.2 Lockwashers. Lockwashers shall be as specified on Drawing SC-A-46420 or shall conform to MS35338.

3.4.1.3 Anti-seize compound. Anti-seize compound shall be used to prevent seizing of threads where the threaded surface is aluminum or magnesium alloy. The compound shall be made of equal parts by weight of petroleum and fine (200 mesh) zinc dust.

3.4.1.4 Solder and soldering. Solder conforming to QQ-S-571 shall be applied in accordance with requirement 5 of MIL-STD-454.

3.5 Construction.

3.5.1 Aneroid mechanism. The aneroid mechanism shall consist of an evacuated metal cell, sensitive to variations in atmospheric pressure, connected to an indicating pointer by a lever system in such a manner that the motion of the cell, for a given change in atmospheric pressure, will be magnified by the lever system and transmitted to the indicating pointer with a minimum of friction in the moving parts and a minimum of distortion other than that which may be intentionally introduced to obtain desirable calibration characteristics.

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3.5.2 Pressure sensitive cell. The structure of the cell shall be such that if soldered joints are used, they shall be subjected to a minimum of binding and shearing stresses; or, if this condition is not accomplished, the area of the soldered joints shall be sufficiently large to ensure that coldflow of the solder under strain will not be sufficient to cause an appreciable deformation. Shearing stresses in limited areas of the flexible surfaces of the cell shall be eliminated as nearly as possible. To accomplish this, the structure of the cell shall be such that the force of any balancing spring that may be used will be symmetrically applied to the flexible surfaces of the cell and will be, as nearly as possible, uniformly distributed over these surfaces.

3.5.3 Syphon-type cell. If the syphon-type cell is used, the structure shall be such that the internal balancing spring will maintain the upper and lower surfaces of the cell sensibly parallel throughout the entire pressure range of the cell. Cells in which the pressure of the spring is unevenly distributed, causing the cells to lean or tip to one side, shall not be acceptable. The cell shall be so constructed that the balancing spring cannot change its position relative to the surfaces on which it bears, as the cell expands or contracts. The balancing spring shall not come in contact with the flexible surfaces of the cell at any place throughout the pressure range of the instrument, whatever the type of the cell structure. The movement of the cell as it expands and contracts with uniformly varying pressure shall be smooth and uniform throughout the pressure limits of the cell.

3.5.4 Lever system.

3.5.4.1 Design. The design of the lever system shall be such that a total pressure range of not less than 320 mb will be indicated. At no point in this range shall a pressure difference of 3 mb be indicated by a linear distance of less than 0.25 inch (6.35mm) of the arc of the scale on the indicating dial. No more than two complete revolutions of the pointer shall be provided.

3.5.4.2 Pressure cell mounting. The pressure cell shall be mounted with one end of the cell attached to the mounting frame so that its position relative to the frame can be altered only by intentional adjustment. The other end of the pressure cell shall be attached to the lever mechanism and shall be free to move in response to changes in the dimensions of the cell caused by variations in atmospheric pressure. The lever system and pressure-sensitive cell shall be so constructed that a linear calibration of the indicating dial may be used.

3.5.4.3 Position compensating. The lever system shall be accurately balanced or counter-balanced, or maintained under sufficient spring tension, to insure that unbalanced weights will not cause shifts in the position of the indicating pointer greater than 0.3 mb on the pressure scale when the dial is placed in a vertical position and then in a horizontal position, nor greater than 0.5 mb when the dial is in a vertical position and rotated 360°.

3.5.4.4 Friction, looseness, and backlash. The construction of the lever system shall produce a mechanism with a minimum of friction, looseness of joints, and backlash. The indicating pointer shaft and all other shafts in the system shall have jeweled bearings throughout. The number of moving parts in the system shall be kept to the least number possible. Designs employing link chains shall not be accepted. The gear on the pointer shaft and the sector gear of the lever system shall be under tension to reduce backlash.

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3.5.5 Stop. A stop shall be provided to prevent the linkage from becoming disengaged when the barometer is exposed to pressure beyond the range specified in 3.1. If the stop is to bear on the pressure sensitive cell it shall be positioned in such a manner that no unbalanced strains will be established in the cell when it is subjected to pressures beyond the range of the stop.

3.5.6 Adjustment mechanisms. Means shall be provided to enable the indicating pointer to be set to read current atmospheric pressure as indicated by a standard mercurial barometer. Construction shall be such that the adjustment cannot be changed unintentionally or by severe vibration. One adjustment shall be provided for large scale adjustments, another shall be provided for accurately making very small changes in the indication of the instrument. The two adjustments may be incorporated into the same construction. The large scale adjustment shall be capable of being made without in any way varying the calibration characteristics of the instrument. This requirement shall also apply where the two adjustments are incorporated in the same construction.

3.5.6.1 Adjustment. Adjustment to the mechanism shall be made with a screwdriver from outside the instrument. The slotted screwhead shall be at least large enough to accommodate a 0.219 inch (5.56mm) wide screwdriver blade. The width of the slot shall be not less than 0.03 inch (.76mm). Construction shall be such that access to the adjusting mechanism or adjustment thereof cannot damage other parts of the instrument. Adjustment shall be made through either the side or the front of the instrument case. No part of the adjusting means shall project outside the instrument case. Means which require adjustments to be made through the back of the instrument case, or require the position of the instrument to be changed during the adjustment, shall not be acceptable.

3.5.7 Indicating shaft. The indicating pointer shaft shall be mounted in the lever system, perpendicular to the plane of the indicating dial. The shaft shall project above the surface of the dial to allow sufficient clearance between the dial and the pointer.

3.5.7.1 Indicating pointer. The indicating pointer shall be straight, light in weight and accurately balanced with reference to its point of attachment on the pointer shaft. The pointer shall be made of light colored metal that will contrast with the finish of the sighting edge. The part of the pointer that projects over the mirror and the graduated scales of the indicating dial shall be not greater than .006 inch (.15mm) wide in a cross section parallel to the dial surface. The pointer shall be constructed to prevent permanent deformation of shape by severe jarring of the instrument. The fit of the pointer on the shaft shall not be altered by repeated removal and replacement of the pointer. The pointer shall be removable without the use of special tools. The sighting edge of the pointer that projects over the mirror and the graduated portion of the dial shall be finished dull black. The sides of the pointer that are perpendicular to the dial shall not be painted. There shall be a clearance of .094 inch (2.39mm) between the pointer and the dial.

3.5.8 Mounting frame assembly. The entire mechanism, the indicating dial and a stop for the pressure cell shall be mounted on a frame. The indicating dial shall be an integral part of the mounting frame if desired, provided the adjustment means specified in 3.5.6 do not require a rotatable dial. The mounting frame structure shall be

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sufficiently rugged so that all parts of the aneroid mechanism will be maintained in alignment, without changes in calibration greater than the errors specified in table I after the instrument has been subjectd to the vibration test specified in 4.6.6.4. The entire mounting frame may be suspended from the instrument case in such a manner that the aneroid mechanism shall not receive the the full force of the strains and vibrations to which the case is subjected. This form of suspension, however, shall be sufficiently stiff to keep the mounting frame substantially centered in the instrument case when the instrument is at rear and suspended in its specified reading position. If a suspension structure is employed, the movement of the mounting frame in this suspension shall be not greater than .188 inch (4.78mm) by means of not less than three stops spaced uniformly around the frame. The use of a spring suspension shall be acceptable, provided the springs do not break or become disengaged during the vibration test.

3.5.9 Indicating dial. The indicating dial, or at least the base for the dial, shall be noncorroding metal plate or a suitable plastic. The face of the dial shall have a matte finish. It shall be uniformly colored throughout in a color contrasting well with the black graduation lines. The dial shall be as large as the instrument case cover and the aneroid mechanism will permit. Dials less than 5 inches (127mm) in diameter shall not be acceptable. The diameter of all scales shall be the maximum permitted by the size of the dial. A mirror ring or mirror insert shall be provided on the dial surface, concentric with the scales. The dial shall be marked to indicate in what position the instrument shall be used for most accurate results as shown on DWG SC-C-20682. The dials shall be either hand-calibrated for each individual instrument or all shall be made from the same master dial. The dial shall be in accordance with DWG SC-C-20582 except as otherwise specified herein.

3.5.9.1 Lines. All lines shall be the same length as shown on DWG SC-C-20582 and the width of the lines shall be not greater than .007 inch (.18mm). The letters, figures, and the spacing of scales related to each other shall be the same size as shown on the above drawing. The spacing between graduation lines on the scales as shown on DWG SC-C-20582 is for information purposes only. All graduation lines shall be portions of radii extending from a common point on the axis of the needle shaft, and are to be of uniform width.

3.5.10 Instrument case. The case shall be wood, metal or molded plastic, with general size, shape and appearance in accordance with DWG SM-C-282049. The length of sides shall be not greater than 7 inches (177.8mm) and the height shall be not greater than 4 inches (101.6mm). If the case is wood constructed, all corners of the case shall be reinforced with metal. The rear corners may be reinforced by the hinge bars. The aneroid mechanism shall be attached to the case in a manner that changes in alignment in the case, due to warping or shrinking, can impose no strains on the aneroid mechanism. If the case is metal constructed, the metal shall be noncorroding, and shall be not less than 0.040 inch (1.016mm) thick. If such material is not available, a ferrous metal shall be substituted provided it is protected by "Parkerizing" or a similar process. If the case is plastic constructed, the material shall be in accordance with MIL-M-14, type CFI-5. The instrument case shall be splash-proof, but need not be air-tight.

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3.5.10.1 Window. A window slightly larger in size than the indicating dial shall be attached to the case as a cover for the dial. The window shall be such that all figures and graduations on the dial will be visible when viewed from any point directly above the window. The window material shall be plastic conforming to MIL-P-5425, finish A. The window or the window mounting shall be removable and replaceable without the use of special tools. The clearance between the window and the indicating pointer shall be not less than .125 inch (3.18mm).

3.5.10.2 Lid. A hinged lid shall be provided over the window. The lid shall be a removable type. The hinge on the lid shall be as long as the shape of the lid and case shall permit. The side of the lid opposite the hinge shall have a positive catch.

3.5.10.3 Pendant ring. A pendant ring shall be attached to the instrument case to hang the barometer in a vertical position. The ring shall fold flush with the case when not in service.

3.5.11 Correction chart. A single chart shall be attached to the instrument case. The chart material shall be a good grade of white book paper. The information on the chart shall be as specified on DWG SM-C-282049 and 3.5.11.1 through 3.5.11.4.

3.5.11.1 Calibration graph. A graph showing differences in calibration due to changes in temperature shall be furnished for each instrument, from which it shall be possible to obtain the correction to be applied at any temperature within the operating limits. The data for this curve shall be obtained from each barometer individually; data giving the average performance for a group of instruments will be unacceptable.

3.5.11.2 Scale error graph. A graph shall be furnished for each instrument showing the scale error of that instrument, from which it shall be possible to evaluate the correction to be applied to any indicated pressure at room temperature over the entire range of the instrument.

3.5.11.3 Scale. A scale shall be furnished for each instrument, from which it shall be possible to convert pressure values in millibars (mb) to the equivalent values in inches, to the nearest 0.003 inch (.08mm) of mercury over the complete range of the instrument.

3.5.11.4 Nomograph. A nomograph shall be furnished for converting differences in pressure (millibars) to differences in elevation (feet), corrected for temperature of the intervening air column. The nomograph shall cover the pressure range of the barometer and a temperature range from 120°F (49°C) to -40°F (-40°C). It shall be prepared in accordance with the Smithsonian meteorological tables, 6th revised edition.

3.5.12 Scale calibration. The barometer calibration shall be checked at ambient room temperature, 77 ±7°F (25 ±3.9°C) against a standard manometer. The allowable calibration error shall be as specified in table I.

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Table I. Allowable calibration error.

Range of pressures (millibars)	Max. Error at any point 1/ 2/ (millibars)
1065 - 926	0.6
925 - 851	1.0
850 - 745	1.6

1/ The change in error between adjacent test points shall be not greater than 0.6 mb throughout the test range.

2/ At the same pressure, the difference in readings between data taken at increasing and at decreasing pressures shall be not greater than 0.8 mb at any point in the test range.

3.5.13 Repeatability. The barometer calibration shall be checked against a standard manometer at a temperature differing by not greater than 5°F (2.8°C) from that which prevails in 3.5.12. The calibration shall not differ at any point from the corresponding data obtained in 3.5.12 by greater than 0.4 mb.

3.5.14 Accuracy comparison. The average of the error obtained by the tests specified in 4.6.5.2 and 4.6.5.2.1 shall be computed and shall not differ by greater than 0.3 mb from those shown on the correction curve as specified in 3.5.11.2.

3.5.15 Change in error at 120°F (49°C). The errors of the barometer at 120°F (49°C) shall not differ from the average of the errors found in the tests specified in 4.6.5.2 and 4.6.5.2.1 by greater than 0.7 mb at any point throughout the range.

3.5.16 Change in error at 30°F (-1°C). The errors of the barometer at 30°F (-1°C) shall not differ from the average of the errors found in the tests specified in 4.6.5.2 and 4.6.5.2.1 by greater than 1.0 mb at any point throughout the range.

3.5.17 Carrying case. The barometer carrying case CY-1059( )/UM shall be constructed in accordance with DWG SC-DL-282023.

3.6 Maintainability. The barometer shall be such that adjustment, maintenance, repair and replacement of parts are easily accessible. Tools shall be combination type tools.

3.7 Service conditions. The barometer shall meet the following service conditions (see 4.6.6).

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3.7.1 Altitude. The barometer shall be operable without degradation in specified performance at altitudes from 1,385 feet below sea level to 8,267 feet above sea level, corresponding to atmospheric pressures of 1065 millibars and 745 millibars, respectively, and shall withstand air transportation at altitudes up to 40,000 feet (12,192m) above sea level (see 4.6.6.1).

3.7.2 High temperature. The barometer shall be operable without deterioration damage or degradation in specified performance at ambient air temperature as high as +120°F (49°C) and shall be capable of withstanding, without degradation of specified performance, long periods of storage at temperatures as high as +160°F (71.1°C) (see 4.6.6.2).

3.7.3 Low temperature. The barometer shall be operable without deterioration damage or degradation in specified performance at air temperatures as low as 30°F (-1°C), and shall be capable of withstanding, without degradation of specified performance, long periods of storage at temperature as low as -60°F (-51°C) (see 4.6.6.3).

3.7.4 Humidity. The barometer shall be operable without degradation in specified performance and shall sustain no physical damage, during and after prolonged exposure to extreme humidity levels as encountered in tropical areas (see 4.6.6.4).

3.7.5 Fungus. After the fungus test of 4.6.6.5, the barometer shall be visually examined. The barometer shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area and no more than six unrelated minuted colonies with mycelial development in areas only in other than critical circuit portions, such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. In addition, the barometer shall operate without degradation in specified performance after exposure to the test conditions of 4.6.6.5. Materials used in the manufacture, fabrication, construction and treatment of the barometer shall comply with Requirement 4 of MIL-STD-454.

3.7.6 Dust. The barometer shall be operale without degradation in specified performance after being subjected to the dust test of 4.6.6.6.

3.7.7 Vibration. The barometer shall operate within specified performance after being subjected to vibration test specified in 4.6.6.7.

3.7.8 Shock. The barometer shall operate within specified performance after being subjected to the method and condition of the shock test specified in 4.6.6.8.

3.7.9 Bench-handling. The barometer shall withstand the shock as encountered during servicing. The equipment shall withstand this test removed from its normal enclosure (see 4.6.6.9).

3.7.10 Calibration. After completion of the vibration tests in 4.6.6.7, the barometer shall be checked against a standard manometer at ambient room temperature, 77 ±7°F (25 ±3.9°C), as the pressure is changed in accordance with table I.

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3.8 Hysteresis. When tested as specified in 4.6.7 the difference in error before decreasing the pressure and the error 8 hours after return to ambient pressure shall be not greater than 0.3 mb.

3.9 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance.

3.10 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be forest green semi-gloss enamel.

3.11 Marking. Marking shall be in accordance with MIL-M-13231 and DWG SC-C-20582.

3.11.1 Serial numbers. The equipment shall be serial numbered.

3.12 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspections (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article units. When specified in the contract, the contractor shall furnish first article units of the complete barometer (see 6.3).

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for group A, group B and group C. Failure of any inspection shall be cause for rejection.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Material	3.3
Finish	3.10
Marking	3.11

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Group A inspection shall be as specified in table II. Sampling shall be in accordance with MIL-STD-105, inspection level II. Acceptable quality level (AQL) shall be as specified in table II.

Table II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Visual and mechanical	3.12	4.6.2	1.0	4.0
Barometer	3.5	4.6.2		
Carrying Case CY-1059( )/UM	3.5.17	4.6.2		
Position compensating	3.5.4.3	4.6.3	1.0 for combined group	1/
Friction, looseness and backlash	3.5.4.4	4.6.4		
Scale calibration	3.5.12	4.6.5.2		
Repeatability	3.5.13	4.6.5.2.1		
Accuracy comparison	3.5.14	4.6.5.3		
Change in error at 120°F (49°C)	3.5.15	4.6.5.4		
Change in error at 30°F (-1°C)	3.5.16	4.6.5.5		

1/ All defects are considered major.

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4.5.1.1 Order of inspection. Group A inspection shall be performed in any order that is satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall be as specified in table III.

Table III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent (defective)
Interchangeability	3.9	4.6.8	6.5

4.5.2.1 Group B sampling plan. Sampling shall be in accordance with MIL-STD-105, inspection level S-4 for normal and tightened inspection, and level S-3 for reduced inspection.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV.

Table IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Altitude	3.7.1	4.6.6.1
High temperature	3.7.2	4.6.6.2
Low temperature	3.7.3	4.6.6.3
Humidity	3.7.4	4.6.6.4
Fungus	3.7.5	4.6.6.5
Dust	3.7.6	4.6.6.6
Vibration	3.7.7	4.6.6.7
Shock	3.7.8	4.6.6.8
Bench handling	3.7.9	4.6.6.9
Calibration	3.7.10	4.6.6.10
Hysterisis error	3.8	4.6.7

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4.5.3.1 Group C sampling plan. Two sample units shall be selected at random for all inspections in table IV from every 200 or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Group C failures. Actions required relative to group C failures shall be as specified (see 6.2).

4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Methods of inspection.

4.6.1 Inspection standards. Prior to submission of samples as specified in 3.2, the contractor shall submit a complete description of the prime standards and the test standards he proposes to use in the inspection of barometers.

4.6.1.1 Prime standards. The prime standard shall be a mercurial manometer which has been certified by the National Bureau of Standards. Three certified copies of the most recent calibration shall be submitted for the approval of the contracting officer.

4.6.1.2 Test standards. The test standards shall be mercurial or aneroid manometers. These manometers, when compared with the prime standard on two successive runs, shall be able to repeat readings at the same points throughout the range within 0.2 mb. Three certified copies of the most recent comparative runs shall be submitted for approval by the contracting officer.

4.6.1.3 Rechecking aneroid inspection standards. The calibration of the aneroid standard against the prime standard shall be rechecked throughout the test range for which they will be used, before making any inspection and at intervals not exceeding one month during the time it is used for inspection. The errors found in the aneroid standard shall be repeated to within 0.2 mb throughout the range, at any time the aneroid standard is rechecked against the mercurial standards.

4.6.2 Visual and mechanical inspection. The barometer including Case CY-1059( ) /UM shall be inspected for conformance to 3.12 and for those items covered by subsidiary documents in 4.4. The defects shall be classified as major or minor in accordance with table V.

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Table V. Classification of visual and mechanical defects.

	Defects
Major	Fastener missing, damaged, or not seated. Fastener improperly locked. Lockwasher missing or broken. Burred screw, with slot useless. Any foreign object that could cause jamming of a mechanism, damage to gear teeth or other driving mechanism. Parts missing, inoperative, improperly assembled, or defective.
Minor	Fastener improperly locked. Lockwasher missing or broken. Flatwasher missing.

4.6.3 Position compensating. Readings shall be taken of the instrument when placed so that the dial is in a vertical position and then in a horizontal position. The difference in indication between these two positions when taken at the same elevation shall be not greater than 0.3 mb on the pressure scale. With the dial in a vertical position, the instrument shall be rotated slowly through 360° in any plane, at the same elevation. The difference in indication between any two positions shall be not greater than 0.5 mb. Nonconformance to 3.5.4.3 shall constitute failure of this test.

4.6.4 Friction, looseness and backlash. The test for friction, looseness and backlash in the aneroid mechanism shall be as follows:

a. The barometer shall be hand held and read. Without jarring, or in any way rotating the dial, the barometer shall be carefully lowered a distance of 36 inches (914mm). There shall be a perceptible increase in the indication of the pointer. This test shall be repeated except that the barometer shall now be raised 36 inches (914mm). There shall be a perceptible decrease in pointer indication.

b. The barometer shall be subjected to a pressure change of 0.3 mb. The indication of the pointer shall change not less than 0.2 mb without vibrating the instrument.

c. The indicating pointer shall be carefully displaced by hand a distance on the dial equivalent to 15 mb of pressure, carefully allowed to return to its original position without jarring or tapping the pointer, and read. The same procedure shall be repeated, moving the pointer in the opposite direction. The difference between readings shall be not greater than 0.2 mb.

d. The movement of the indicating pointer during the tests specified in 4.6.7 shall be carefully noted. When the pressure is changed uniformly there will be no evidence of uneven motion or hanging of the pointer at any point on the dial. Irregular motion of indicating pointer at any point on the scale, with uniformly varying pressure shall constitute failure of this test.

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4.6.5 Scale calibration and temperature tests.

4.6.5.1 Test procedure.

a. The instrument shall not be subjected to pressure variations greater than 1 inch (25.4mm) of mercury for a period of 15 hours preceding each test.

b. For a period of not less than 1 hour preceding the the test, the instrument temperature shall be maintained within ±5°F (±2.8°C) of the temperature at which the test is to be made. During the test, the allowable temperature tolerance shall be ±2°F (±1.1°C).

c. Every reasonable effort shall be made to reduce leakage to a minimum in test equipment. When leakage is present the tendency of this leakage shall be adjusted by means of suitable control valves, or other means, so that it is very small and in the same direction as the change in the pressure being progressively maintained at that time in the test. When tests are being made at progressively lower pressures, the tendency of any pressure change occurring at the time data is taken shall also be decreasing, and vice versa.

d. In test requiring the instrument to be subjected to change in pressure, the pressure shall first be reduced from current atmospheric pressure to the low pressure limit of the test, then increased to the upper pressure limit of the test, and finally, reduced to atmospheric pressure. Unless otherwise required, the lower pressure limit of all tests shall be not greather than 745 mb and the upper limit not less than 1065 mb. Test data shall be taken at both increasing and decreasing pressures.

e. In making the tests at different pressures, unless otherwise required, test points shall be taken at not greather than 25 mb intervals. Test points at which data are taken shall have the same pressure value, for both increasing and decreasing pressures, to within 1.0 mb in any one test cycle. In tests made to determine changes in reading due to changes in temperature, data shall be taken at the same pressure values (±0.3 mb) at each temperature.

f. In making the tests at different pressures, the rate of pressure change between test points shall be not more than 50 mb/min. The pressure shall be held approximately constant for not less than 5 minutes prior to obtaining test data.

g. Where reference is made to the error permitted in a test, this error shall be defined as the difference between the indication of barometer and the corrected indication of the standard manometer when data from both instruments are taken simultaneously. In noting the indication of barometer, the instrument may be lightly tapped or vibrated; then the position of the indicating needle shall be estimated as closely as possible. The indications of the test manometer shall be carefully determined to the nearest 0.05 mb or thousandth-inch.

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4.6.5.2 Scale calibration. The instrument shall be maintained at the ambient room temperature for this test, 77 ± 7°F (25 ± 3.9°C). The pressure shall be changed in accordance with 4.6.5.1 and comparative readings made between the test barometer and the standard manometer. Nonconformance to 3.5.12 shall constitute failure of this test.

4.6.5.2.1 Repeatability. The test specified in 4.6.5.2 shall be repeated, taking data at the same test points as before and at a temperature differing by not greater than 5°F (2.8°C) from that which prevailed in the original test. The data obtained on this second test shall be compared at any point with the corresponding data obtained in the test specified in 4.6.5.2. Nonconformance to 3.5.13 shall constitute failure of this test.

4.6.5.3 Accuracy comparison. Average the error values obtained in 4.6.5.2 and 4.6.5.2.1. Nonconformance to 3.5.14 shall constitute failure of this test.

4.6.5.4 Change in error at 120°F (49°C). This test shall be conducted as specified in 4.6.5.1. The errors of the barometer at this temperature shall not differ from the average of the errors found in the tests specified in 4.6.5.2 and 4.6.5.2.1 by more than 0.7 mb at any point throughout the range. The errors found in this test shall not differ from those shown on the temperature correction curve specified in 3.5.11.1 by more than 0.5 mb. Evidence of barometer errors averaging more than 0.7 mb or errors differing from those shown on the temperature correction curve by more than 0.5 mb shall constitute failure of this test.

4.6.5.5 Change in error at 30°F (-1°C). This test shall be conducted as specified in 4.6.5.1. The errors of the barometer at this temperature shall not differ from the average of the errors found in the tests specified in 4.6.5.2 and 4.6.5.2.1 by more than 1.0 mb at any point throughout the range. The errors found in this test shall not differ from those shown on the temperature correction curve for the instrument as specified in 3.5.11.1 by more than 0.5 mb. Evidence of barometer errors averaging more than 1.0 mb or errors differing from those shown on the temperature correction curve by more than 0.5 mb shall constitute failure of this test.

4.6.6 Service conditions. The service condition tests shall be performed as detailed below. Each barometer subjected to these tests shall have passed group A tests and shall be resubmitted to and pass these group A tests after completion of all service condition tests.

4.6.6.1 Altitude.

4.6.6.1.1 Storage and transportation. The barometer shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. The test shall be performed after the barometer has been resting for a period of 15 hours, during which it was not subjected to pressure variation in excess of 25.4 mm (1 inch) Hg. In Step 2, the chamber pressure shall be adjusted to 141 mm of Hg (5.56 inches of Hg), corresponding to an altitude of 40,000 feet (12,192 m) above sea level, at a rate not to exceed 2000 fpm. In Step 3, maintain the chamber pressure for 4 hours. In Step 5, conduct an operational checkout 24 hours after returning to ambient pressure. The instrument error shall not exceed 0.3 mb in both the pre-test and post-test checks, when compared with a standard barometer.

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4.6.6.1.2 Operation. After completion of the shock and vibration tests of 4.6.6.7, 4.6.6.8 and 4.6.6.9, the barometer shall be subjected to the test of MIL-STD-810, Method 500.2, Procedure II. In Step 2, adjust the chamber pressure to 745 mb, corresponding to an altitude of 8,267 feet (2519 m). Vary the chamber pressure over the range given in table I of this specification, at a rate not to exceed 2000 fpm. The maximum error in the instrument reading during the pressure variation shall be as given in table I.

4.6.6.2 High temperature. The barometer shall be subjected to the test of MIL-STD-810D, Method 501.2, Procedure II for cycling exposure. In Step 12, raise the internal chamber temperature to +120°F (+49°C). Raise the internal chamber temperature to +160°F (+71°C) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +120°F (+49°C) and maintain this temperature until temperature stablization of the test item is reached. Operate the test item until the item is stabilized.

4.6.6.3 Low temperature. The barometer shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in Step 2 to -60°F (-51°C) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In Step 5 of Procedure I, perform Procedure II - Operation (paragraph 3.2 of Method 502.2). The chamber temperature in Step 1 of Procedure II shall be adjusted to +30°F (-1°C).

4.6.6.4 Humidity. The barometer shall be subjected to the test of MILSTD810D, Method 507.2, Procedure III. Sealed equipment shall also be subjected to Method 507.2, Procedure II. In Step 4 of Procedure II, cycle 5 of Table 507.2-I shall be used.

4.6.6.5 Fungus. The barometer in the assembled-and-ready-for-delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the barometer or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area and no more than six unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, printed circuit boards, etc. The barometer shall fail if it does not meet operational performance criteria or if it shows more than the growth specified above.

4.6.6.6 Dust. The barometer shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. In Step 2 and Step 5, the air velocity shall be 300 ± 200 feet per minute. In Step 5, the air temperature shall be adjusted to +145°F (+63°C). The barometer shall be reoriented periodically to expose all of its sides to the dust stream. The barometer shall be operated before Step 1, during Step 8, and during Step 11.

4.6.6.7 Vibration. The barometer shall be subjected to the test of MIL-STD-810, Method 514.3 for Category 8 - ground mobile. The barometer shall be held to the vibration generator by straps or by any other convenient means. The vibration environment shall be simulated using test schedules given in MIL-STD-810D, figures 514.3-7, 514.3-8, and 514.3-9. The duration of testing shall be 240 minutes per axis, for a total test time of 12 hours. The test item shall be reoriented such that each of its three major axes are exposed to vibration for 80 minutes during each 240 minute period.

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In addition, isolated barometer shall also be vibrated with the isolators removed or blocked out as follows: A straight line spectrum from 5-200 Hz such that the Acceleration Spectral Density (ASD) at 5 Hz = 0.1 g^2/Hz and the ASD at 200 Hz = 0.0025 g^2/Hz. This level is applicable to all three directions of vibration. The barometer shall be vibrated for 60 minutes in each of three mutually perpendicular axes, for a total of 3 hours. Prior to and following these tests, the barometer shall be operated in accordance with MIL-STD-810D, general test performance guidance, paragraph 4.5. The barometer shall not have any degradation in performance and shall not suffer any mechanical damage.

4.6.6.8 Shock. The barometer shall be subjected to the test of MIL-STD-810D, Method 516.3, procedure I. The test item shall be held to the shock machine by straps or by any other convenient means. The shock pulse shall be 20g peak for a nominal duration of 11 msec, in accordance with figure 514.3-4.

4.6.6.9 Bench handling. The barometer shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI.

4.6.6.10 Calibration. After completion of all environment tests, the barometer shall be checked as indicated in 3.7.10. Nonconformance to 3.7 shall constitute failure of these tests.

4.6.7 Hysteresis error. The barometer shall not be subjected to pressure variations in excess of 1 inch (25.4mm) of mercury for a period of 15 hours preceding the test. The barometer shall be compared with a standard barometer, at room pressure, and shall then be subjected to a pressure of approximately 745 mb for a period of not less than 15 hours. During this interval the pressure shall not vary by more than 50 mb. The temperature to which the barometer is exposed shall be room temperature. At the end of this interval, the pressure shall be increased to ambient in not more than 5 minutes, and the barometer read as soon as possible. The difference between the error before and the error after this test shall not exceed 1.2 mb. After a period of 8 hours this difference in error shall be recorded. Nonconformance to 3.8 shall constitute failure of this test.

4.6.8 Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies.
b. Dimensions of cavities.
c. Location of hinges and fasteners on separable parts or assemblies which mate.
d. Size and form of special threads.

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4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table VI "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

Table VI. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

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5. PACKING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JR-1-1-00-00-NS-X-ED-0-00-A.

5.1.2 Level B. JR-1-1-00-00-NS-X-ED-0-00-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Barometer, Aneroid ML-102( ) is used for measuring atmospheric pressures from 745 to 1065 mb.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and quantity of first article units (see 3.2).

c. Group C failure action (see 4.5.3.2).

d. Level A or B preservation and packing (see section 5).

e. When first article inspection rough handling tests are not required.

f. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of the quantity of first article units specified in the contract. The contracting officer should include specific instructions in acquisition documents for examination, test, and approval of the first article.

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: ML-102W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Air Force - 99

Preparing activity:
Army - ER
(Project 6660-0078)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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